INCH-POUND
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MILITARY SPECIFICATION

DEHYDRATOR, DESICCANT, HD-186/ASQ

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for one type of dehydrator designated dehydrator, desiccant, HD-186/ASQ.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2).

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation and Packaging, Methods of

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRFW, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4440

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards: Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. Air Force

53C29650	Dehydrator, Desiccant, HD-186/ASQ

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The dehydrator shall consist of all the components shown on Drawing 53C29650 and associated drawings.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Unless otherwise specified, materials shall be as specified on Drawing 53C29650 and associated drawings.

3.4.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.4.2 Non-magnetic materials. Non-magnetic materials shall be used for all parts of the dehydrator.

3.4.3 Castings. Castings shall be of high quality, clean, sound and free from blow holes, porosity and cracks.

3.5 Design and construction. The dehydrator shall be in accordance with Drawing 53C29650 and associated drawings.

3.6 Immobilization of desiccant. Extreme care shall be taken to ascertain that the desiccant particles are properly immobilized.

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3.7 Flow rate. The dehydrator shall have a minimum flow rate of 400 cubic inches of air per minute at standard atmospheric pressure (29.92 inches Hg) with a maximum pressure drop of three inches of mercury.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.9 Nameplate. The dehydrator shall be provided with identification markings in accordance with MIL-STD-130.

3.10 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the dehydrator, they shall be held or limited accordingly.

3.11 Workmanship. The dehydrator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, machine screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Testing (4.3).

b. Quality Conformance Testing (4.4).

4.3 First article testing.

4.3.1 Test samples. The first article test samples shall consist of three complete dehydrator assemblies.

4.3.2 First article tests. The first article tests shall consist of all the tests described under 4.6 and 4.7.

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4.4 Quality conformance tests. Tests shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.4.1 Individual tests. Each dehydrator unit shall be subjected to the following tests as described under 4.6.1 and 4.7.2.

a. Examination of product.

b. Leakage test.

4.4.2 Sampling plans and tests.

4.4.2.1 Lot. A lot shall consist of dehydrators manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.2.2 Sampling plan. Sampling plan test samples shall be selected in accordance with inspection level S-2 of MIL-STD-105 and shall be subjected to the following tests as described under 4.6.2, 4.7.2, 4.7.3, and 4.7.4.

a. Dimensional examination.

b. Leakage test.

c. Pressure drop.

d. Residual moisture test.

4.4.3 Rejection. A sample failing to pass any examination or test shall be cause for the rejection of the lot.

4.5 Test conditions.

4.5.1 Pressure measurements. Pressures shall be recorded in inches of mercury (Hg).

4.5.2 Preparation for test. The dehydrator shall be prepared for testing as required by the specific test instructions.

4.5.3 Observations. During the progress of all tests, the performance of the dehydrator shall be observed to determine the degree of compliance with all requirements specified herein.

4.5.4 Test data. Test data shall be recorded as required under the specific tests.

4.5.4.1 Tolerance. The tolerance on pressure and the temperature readings shall be accurate to within 2 percent of the numerical figure recorded.

4.6 Examinations.

4.6.1 Examination of product. The dehydrator shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, finishes, and marking.

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4.6.2 Dimensional examination. The dehydrator shall be examined to determine dimensional compliance with Drawing 53C29650.

4.7 Test methods.

4.7.1 Vibration. The dehydrator shall be subjected to a vibration test in accordance with Method 514.1, Category (b), Procedure 1, Curve c of MIL-STD-810. Upon completion of this test, the dehydrator shall be subjected to a leakage test in accordance with 4.7.2. Failure to meet the requirements of the leakage test shall be basis for rejection of the dehydrator.

4.7.2 Leakage. The dehydrator shall be temperature stabilized throughout at 80° F or below immediately prior to this test. The dehydrator shall be immersed in water maintained at a minimum temperature of 130° F for a minimum of 5 minutes to check for leakage. A definite continuous forming and breaking away of air bubbles from definite points on the surface of the dehydrator shall be cause for rejection. Presence of bubbles clinging to the surface of the container shall not be considered cause for rejection.

4.7.2.1 Alternate leakage test. In lieu of the leakage test of 4.7.2, the contractor may make an alternate test as follows: One air connection part of the dehydrator assembly shall be plugged and the other one connected to a source of vacuum. The dehydrator shall be evacuated to a pressure of 10 inches Hg absolute and the line to the vacuum source shut off. During a period of 10 seconds, the reading of the pressure in the dehydrator shall not change by more than 0.4 inches Hg.

4.7.3 Pressure drop. Connect a flow meter to an air pump. On the intake side of the pump connect a throttle valve and adjust it so that the flow meter reads 400 cubic inches of air per minute at standard room conditions. Using a mercury manometer, observe the pressure in the set-up between the pump and flow meter, then connect the dehydrator on the intake side preceding the throttle valve and again observe the pressure in the manometer. The pressure drop shall not exceed 3 inches of mercury.

4.7.4 Residual moisture test. Measure 50 milliliters of water into a nonconductive beaker (plastic). Using a quick reacting thermometer with one degree increments, record the temperature of the water. (NOTE: Water temperature shall be above 60° F). Disassemble one end of the dehydrator and measure a 50 milliliter volume of the desiccant. Pour the desiccant into the 50 milliliters of water and stir the desiccant/water mixture with the thermometer for approximately 30 seconds and observe and record maximum temperature attained. Subtract the original temperature of the water from the temperature of the desiccant/water mixture and record. Repeat this procedure 3 times to obtain the average rise in temperature. An average temperature rise of less than 18° F shall be cause for rejection of the dehydrator.

4.8 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine conformance to the requirements of section 5.

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A, B, or C. Each complete dehydrator shall be preserved and packaged in accordance with Method III of MIL-P-116. The item shall be placed in a close-fitting fiberboard box conforming to the requirements of PPP-B-636, Class - Domestic, Grade 125, Style optional. The container shall be closed in accordance with the appendix to the box specification.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Dehydrators, preserved and packaged as specified in 5.1.1, shall be overpacked in a box conforming to PPP-B-621, Class 2, Style optional; or to PPP-B-601, overseas type, style optional, Grade B. Boxes shall be strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Same as level A, except domestic type containers shall be used in lieu of overseas type.

5.2.3 Level C. Dehydrators preserved and packaged in accordance with 5.1.1 shall be overpacked in a box conforming to PPP-B-636, Class - Domestic, style and grade optional. Weight and size shall not exceed the limits of the box specification. Closure shall be in accordance with the appendix to the box specification.

5.3 Marking. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The dehydrator is intended for use in airborne pressurizing sets AN/ASQ-14 and AN/ASQ-15 to dehydrate intake air used for maintaining desired pressure in pressurized electronic equipments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging and level of packing required(see 5.1 and 5.2).

c. Any special marking required (see 5.3).

6.3 Subject term (Key Word) listing.

Dehydrator
Desiccant
Residual Moisture

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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